higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING PROCESSES
NQF LEVEL 4

8 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
- Do not wear jewellery, especially chains, watches, bracelets or earrings.
- Tie back long hair or wear a hair net.
- Button up overalls and sleeves.
- Do not put your hands in the moving parts of machines.
- Never operate a machine that you are not trained and authorized to operate.
- Never sit or lean against machines.
- Make sure that you know where the emergency switches are on all machines.
- Do not smoke in the workshop.
- Do not wear neckties. (Any relevant answer) (Any 4 × 1) (4)

1.2 It prevents the machine from being switched on until maintenance work is finished and the lock has been removed. (2)

[6]

QUESTION 2

2.1
- Make sure that the locking devices are placed in the same way that they were removed.
- Use the correct lubricants and check for leaks.
- Check the rotational direction of the parts.
- Check the alignment of the components where possible. (4 × 2) (8)

2.2
2.2.1 F
2.2.2 D
2.2.3 E
2.2.4 G
2.2.5 B
2.2.6 A

(6 × 1) (6)

2.3
- To see the manufacture's✓ identification marks on the component✓
- To get a good grip✓ on screws, nuts, bolts and keys✓
- To identify marks that on the component✓ before you dismantle it ✓

(3 × 2) (6)

2.4
- Socket set
- Spanner set (both open-ended and ring)
- Torque wrench
- Spark plugs
- Pliers set that includes diagonal cutters, needle nose, slip joint, adjustable joint pliers, combination set
- Vice grip
- Screwdriver (Any relevant answer) (Any 6 × 1) (6)

[26]

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QUESTION 3

3.1
- Never leave the drill press while it is running.
- Switch off the machine off before making any adjustments.
- Feed downward at a sufficient rate to keep the drill cutting.
- Feed with care as the drill breaks through the underside of the workpiece.
- Use a safe working posture. (Any relevant answer) (Any 4 × 1) (4)

3.2
- Inspect component carefully.
- Use direct measuring equipment such as micrometers and vernier calipers.
- Use fixed gauges, such as an electronic gauge which can be a go or no-go range with prescribed dimensions for components. (3)

3.3
- To prevent the machine from rusting
- To restore the machine to an acceptable level of cleanliness
- To ensure that the next person that uses the machine finds it in a good and clean condition
- To extend the machine's life span (4)

[11]

QUESTION 4

4.1
- Check the thickness of the metal.
- Type of metal
- Check the manufacturer's manual.
- Check the amount of gas in the tanks.
- Attach the pressure regulator.
- Check the pressure gauges.
- Check the flashback arrestors. (Any 6 × 1) (6)

4.2
- In mist systems, liquid coolant is mixed✓ with air to form a mist✓.
- The mist coolant is sprayed through a nozzle✓ onto the cutting tool and the work piece.✓
- Coolant and air enter the nozzle✓ through pipes✓.
- Remember to set the liquid flow first✓ and then add air to create mist✓.

(4 × 2) (8)

4.3
4.3.1 False
4.3.2 False
4.3.3 True
4.3.4 True
4.3.5 True
4.3.6 False

(6 × 1) (6)

4.4
- Adjust the lubricant.
- Perform the cutting operation.
- Confirm blade clearances.
- Cut edges to conform to job specifications.
- Check the cutting operation and complete the cut.
- Inspect and give a report regarding the completed components.
- Mount the blade in a correct frame
- Apply the correct cutting speed
- Adjust to the correct cutting stroke (Any 6 × 1) (6)

[26]

QUESTION 5

5.1
- Check that the drawing dimensions✓ of the component are correct.✓
- Check that the tooling will produce the required tolerances✓ and surface finishes.✓
- Check the tool coordinates✓ to make sure that they do not collide during production.✓
- Check whether the cutting time is fast enough✓ and if it is not, speed it up.✓ (4 × 2) (8)

5.2
- X-coordinate is for horizontal movement
- Y-coordinate is for vertical movement
- Z-coordinate is for transverse movement (3)

5.3
- A CNC machine can produce large amounts of work without losing accuracy.
- Unskilled labour can be used.
- They can produce large amounts of work at half the time taken when using an ordinary centre lathe. (3)

5.4
- The cut depth should be correct.
- The cut should not exceed the boundaries of where it should be.
- The tool should be in contact with the material and cutting, not floating above the material.
- The tool should remain within the boundaries of the cutting table.
- The tool should never cut into holding clamps or the sides of the machine.

(Any relevant answer) (Any 3 × 1) (3)

[17]

QUESTION 6

6.1
- Adjust the working pressure on the oxygen and acetylene.
- Open the acetylene valve and adjust it to create a neutral flame.
- Open the oxygen valve and adjust the flame until it is separate from the nozzle.
- To close down always close the acetylene first. This is to make sure no flammable gases remain inside the torch. (4)

6.2
- Electroplating
- Cladding
- Spraying
- Hot-dipping
- Sherardizing (5)

6.3
- Solder wire
- Nuts and bolts
- Rivets or metal pins
- Tin and lead alloys
- Inert gas
- Adhesives
- Studs
- Welding
- Brazing (Any relevant answer) (Any 5 × 1) (5)

[14]

TOTAL: 100

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